

Enrique Sobrados

October 9, 2026

PhD Student & Research Assistant
Computer Science
University of New Mexico
October 9, 2026

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OVERVIEW

I work at the intersection of security research and applied engineering, where I build automated pipelines and AI-driven tools to measure and detect security and privacy issues at scale. My work has helped VPN providers, SMS-delivered URL services, and satellite systems improve their overall security posture. My current research focuses on investigating security misconfigurations that impact the VPN ecosystem at scale.

SKILLS

Offensive Security: Nmap, Burp Suite, Metasploit, Ghidra, Wireshark, Kali
Programming: Python, C/C++, Java, Ruby, JavaScript
Web Development: Angular, React, Spring Boot, Rails

Machine Learning: PyTorch, Keras, TensorFlow, LLMs, AI Agents
Cloud: AD, Azure, SQL Server, Docker, Kubernetes
Languages: English, Spanish, German

EDUCATION

Ph.D. in Computer Science University of New Mexico, Albuquerque, NM, USA Advisor: Dr. Afsah Anwar	2024-Present
M.S. in Computer Science University of New Mexico, Albuquerque, NM, USA	2025
B.S. in Computer Science University of Engineering and Technology, Lima, Peru	2023

PUBLICATIONS

- [1] **Exposing Probe-Resistant VPNs via Protocol Coexistence**
Sobrados, E., Danish, M., Vanlyssel, J., Ensafi, R., and Anwar, A.
IEEE Symposium on Security and Privacy (S&P), 2027.
Identified a weak-link vulnerability in commercial VPNs, where providers co-locate probe-resistant protocols such as WireGuard with probe-sensitive protocols such as IKE on the same server, allowing sensors to fingerprint infrastructure designed to be otherwise undetectable. Using Censys, we measured the server infrastructure of 13 VPN applications—including ProtonVPN, NordVPN, and Surfshark—and found that this weak link affected, on average, 90.1% of their servers, demonstrating how sensors could exploit protocol co-location to map entire VPN infrastructures.
- [2] **MVPNalyzer: An Investigative Framework for the Security & Privacy Audit of Mobile VPNs**
Wang*, W., Ortwein*, A., Sobrados*, E., Stanley, R., Kumar, P., Anwar, A., and Ensafi, R.
Network and Distributed System Security Symposium (NDSS), 2026.
Acceptance rate: 17.89%. (* joint first authors)
Built an investigative framework for large-scale security and privacy audits of mobile VPN ecosystems. Analyzed VPN app backends and infrastructure to identify systemic vulnerabilities, protocol exposures, and privacy leaks. Developed automated tools to measure the efficacy of VPN security claims across diverse network conditions.
- [3] **The Tragedy of Convenience: Cascading User-Data Leakage from SMS-delivered URLs**
Danish, M., Sobrados, E., Kaushik, P., Acharya, B., Saad, M., Mueen, A., Rahaman, S., and Anwar, A.
ACM Conference on Computer and Communications Security (CCS), 2026.
Analyzed 322K URLs from 33M SMS messages. Built an end-to-end pipeline combining data collection, LLM-based PII detection, and manual validation to identify 150+ PII-exposing services. Uncovered weak bearer-token authentication, low-entropy token user enumeration, privileged-action exposures, and account takeover. Responsible disclosures protected 150M+ users.
- [4] **Spychain: Multi-vector supply chain attacks on small satellite systems**
Vanlyssel, J., Sobrados, E., Anwar, R., Roman, G.C., and Anwar, A.
arXiv preprint arXiv:2510.06535, 2025.
Developed a threat model for the SmallSat supply chain, identifying vulnerabilities in COTS hardware and open-source flight software. Proposed a hardware-rooted defense-in-depth framework against supply-chain compromise and unauthorized command execution.
- [5] **Evaluating Commercial VPN Detection Feeds Against Ground Truth**

Sobrados, E., Ensafi, R, and Anwar, A.

In Progress.

An audit of VPN detection and reputation feeds. Using ground-truth VPN entry and exit servers, we measure quality metrics of the commercial VPN detection feeds and quantify the collateral damage to legitimate VPN users.

Target Venue: USENIX Security Symposium, 2027.

PROFESSIONAL EXPERIENCE

Graduate Research Assistant	Present
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University of New Mexico, Albuquerque, NM

- Built **MVPNalyzer**, an extensible mobile-VPN auditing framework for DNS leaks, obfuscation, cleartext traffic, and OpenVPN configurations.
- Disclosed vulnerabilities to 40+ VPN providers, with **6 providers acknowledging vulnerabilities** so far.
- Enumerated infrastructure for **~70,000 VPN servers**; reverse-engineered APIs to map 20+ endpoints.
- Uncovered a **weak-link vulnerability** in popular commercial VPN infrastructure.

Software Engineer	2023
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Meta-Innovaciones S.A.C., Peru

- Developed the front-end for multiple intake desk processes, including administrative litigation.
- Performed secure code reviews and remediated **4 security weaknesses**.
- Automated taxpayer report generation using Jasper Reports + Spring Boot, reducing manual processing effort by over **80%**.
- Assisted in designing a scalable database architecture for multiple system modules, improving data consistency.

Security Analyst Intern	2022
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National Bank, Peru

- Operated VPNs on Palo Alto & Check Point (IPsec/IKEv2) for **~2,000 users**.
- Tuned Forcepoint web filtering, cutting false positives by **15%** while preserving business access.
- Automated ATM key generation/rotation in Bash; standardized ceremonies and saved **~6 hrs/week**.
- Ran vulnerability triage across 2 services; **5 issues** were remediated in **4 weeks**.

TEACHING

Guest Lecturer , University of New Mexico — Intro to Cybersecurity (CS544)	2025
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Teaching Assistant , University of New Mexico — Intro to Cybersecurity (CS544)	2024
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Teaching Assistant , University of New Mexico — Declarative Programming (CS357L)	2023
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Teaching Assistant , Le Wagon — Web Development Bootcamp	2021
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Teaching Assistant , University of Engineering and Technology — Intro to Object-Oriented Programming	2020
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SERVICE

Artifact Evaluation Committee, Network and Distributed System Security Symposium (NDSS)	2027
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Co-Reviewer, Network and Distributed System Security Symposium (NDSS)	2026, 2027
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Co-Reviewer, Annual Computer Security Applications Conference (ACSAC)	2025, 2026
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CERTIFICATIONS

Junior Penetration Tester (eJPTv2)	INE Security
PEN-100, SOC-100, WEB-100	Offensive Security

AWARDS & LEADERSHIP

President, LoboSec (UNM CTF Team)	Present
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NDSS Fellowship , Internet Society	2026
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3rd Place , Lobo Hackathon	2026
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3rd Place & Individual Top Scorer , Alamo CTF	2025
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CS Department Award , University of New Mexico	2024
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1st Place , Tracer Fire CTF, Sandia National Laboratories	2023
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ACSAC Student Conferenceship	2023
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REFERENCES

Afsah Anwar , Assistant Professor	afsah@unm.edu
<i>Department of Computer Science, University of New Mexico</i>	